Lesson 1.6

Constructions

Exact drawings with a compass and a straightedge

The two tools



Compass: draws circles and arcs, copies a length



Straightedge: draws straight lines — **no measuring!**

The two tools



Compass: draws circles and arcs, copies a length



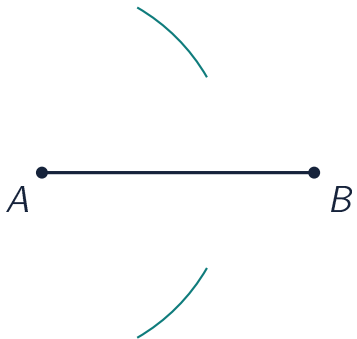
Straightedge: draws straight lines — **no measuring!**

Four basics: copy a segment, copy an angle, **bisect a segment**, **bisect an angle**.

Bisect a segment (find its midpoint)

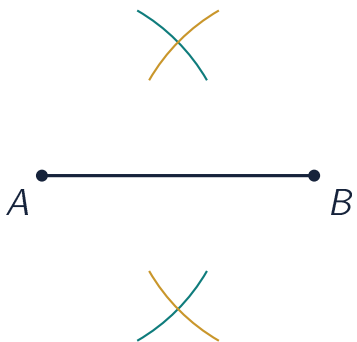


Bisect a segment (find its midpoint)



1. From A: open the compass **more than half** of AB. Arcs above and below.

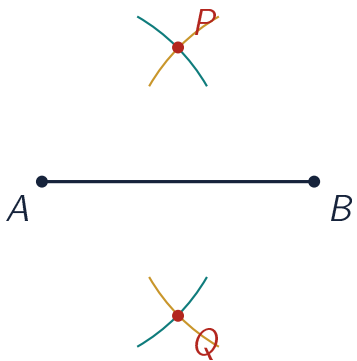
Bisect a segment (find its midpoint)



1. From A: open the compass **more than half** of AB. Arcs above and below.

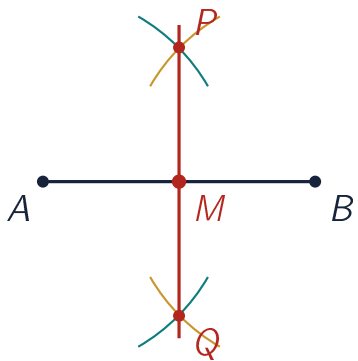
2. From B: the **same** width. Arcs above and below.

Bisect a segment (find its midpoint)



1. From A: open the compass **more than half** of AB. Arcs above and below.
2. From B: the **same** width. Arcs above and below.
3. The arcs cross at P and Q.

Bisect a segment (find its midpoint)

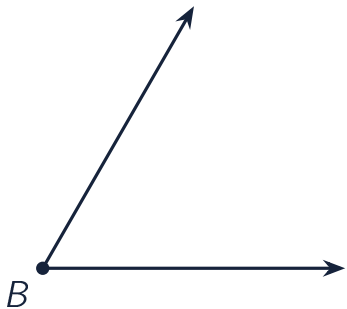


1. From A: open the compass **more than half** of AB. Arcs above and below.
2. From B: the **same** width. Arcs above and below.
3. The arcs cross at P and Q.
4. Line PQ is the **perpendicular bisector**. It meets $\overline{AB}$ at the midpoint M.

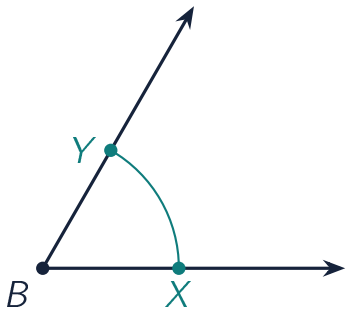
Watch out

Watch out! Keep the compass width the **same** for both arcs, and open it to **more than half** of AB — otherwise the arcs won't cross.

Bisect an angle

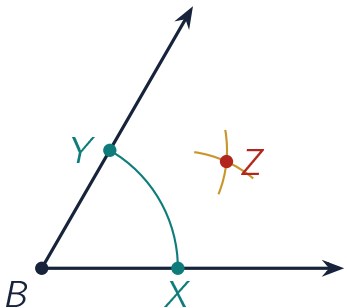


Bisect an angle



1. Arc from the vertex B crosses both sides at X and Y.

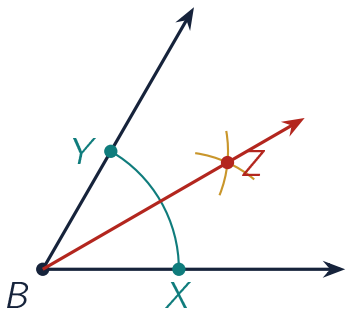
Bisect an angle



1. Arc from the vertex B crosses both sides at X and Y.

2. Same width from X and from Y: the arcs meet at Z.

Bisect an angle

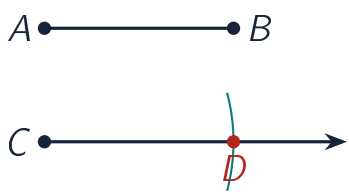


1. Arc from the vertex B crosses both sides at X and Y.

2. Same width from X and from Y: the arcs meet at Z.

3. Ray BZ **bisects** the angle: two equal halves.

Copy a segment



Set the compass to AB. Put the point on C and draw an arc across the ray. Where it crosses is D.

Now $CD = AB$ — copied without measuring.

Your turn! Pause the video and try it.

You want to bisect a segment that is 8 cm long.

Which compass width works: 3 cm or 5 cm?

Your turn! Pause the video and try it.

You want to bisect a segment that is 8 cm long.

Which compass width works: 3 cm or 5 cm?

5 cm — it must be **more than half** of 8 cm, which is 4 cm.

With 3 cm the arcs from A and B never meet.

Remember

- Only a **compass** and a **straightedge** — no measuring.
- Bisect a segment: same width, more than half, from both ends.
- The line through the crossings is the **perpendicular bisector**.

Now practice at your level:

fangmath.com → Geometry → Chapter 1